

Introduction to Data Science

Block 4.1: Data Science

Oliver Lindemann & Yaniv Morgenstern

Erasmus University Rotterdam

What is Data Science?

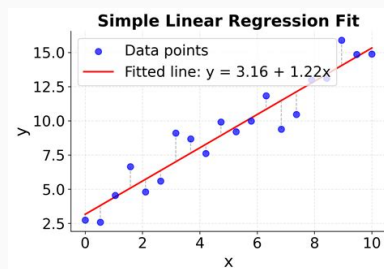
1

2

Data Science

Extracting insights and knowledge from data

- Data Analysis, Statistics
- Modelling of Data
- AI and Machine Learning



3

4

Modelling Data

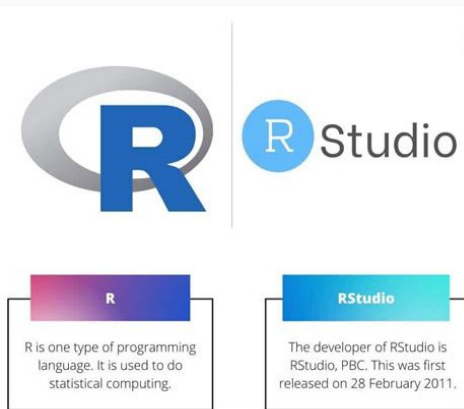
1. Data-Driven Models (*Yaniv*)

- Principles of data fitting
- Neuronal Network Models

1. Theory-Driven Models (*Oliver*)

- Models of Memory: Multinomial Processing Tree Models
- Reaction time models in decision making: Diffusion Models

How to do data science? How to model data?

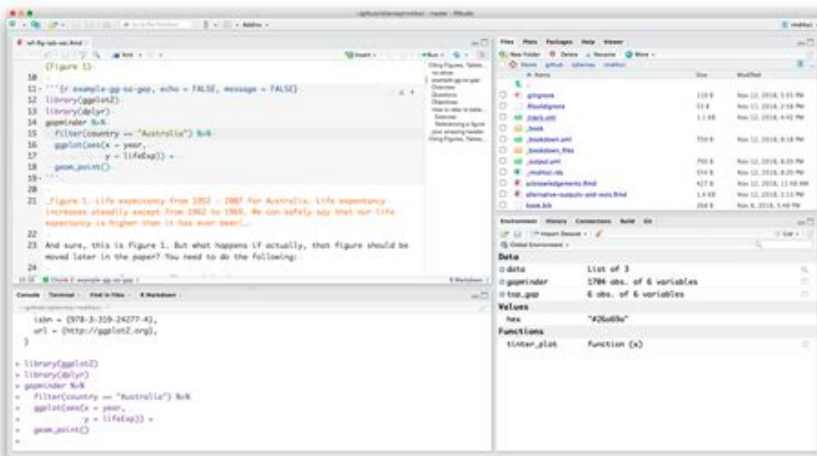


Programming in R (with RStudio)

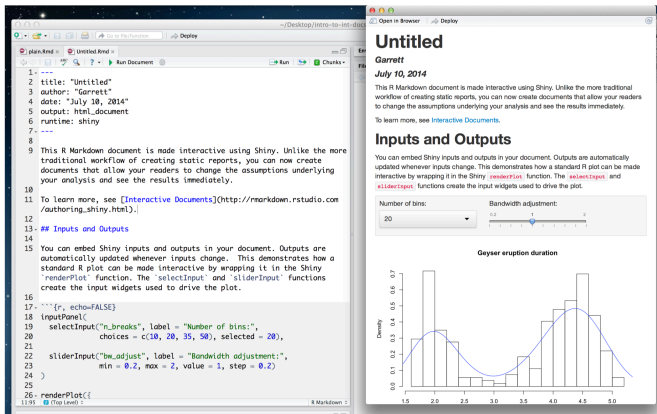
5

RMarkdown

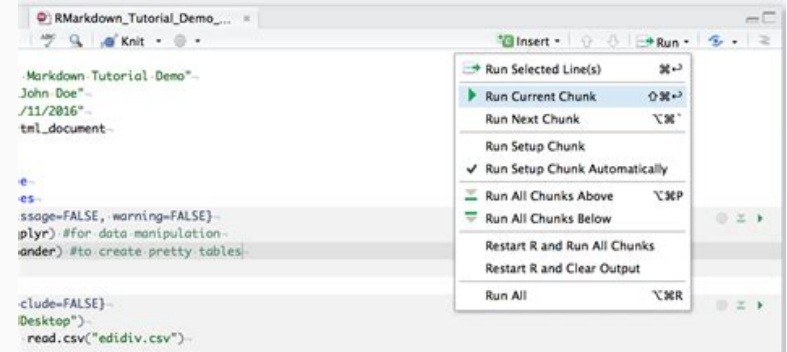
Combines text, code and output in one document



6



Running code of RMarkdown documents (in RStudio)



7

8

Course overview

Date	Topic	Room	Practical (11:00)	Lecture (13:00)
01/09	1	2.02	Introduction 1: R basics	Introduction to Data Science (OL,YM)
08/09	2	2.02	Introduction 2: R basics	Parameter Estimation (YM)
15/09	3	4.02	Parameter Estimation	Neural Networks 1 (YM)
22/09	4	4.02	Neural Networks 1	Neural Networks 2 (YM)
29/09	5	1.12	Neural Networks 2	Cognitive Modelling (OL)
06/10	6	3.14	Cognitive Modelling	Multinomial Processing Tree Models (OL)
13/10	7	2.20	Multinomial Processing Tree Models	Reaction Time Models (OL)
20/10	8	2.02	Reaction Time Models	Q & A Lecture (YM,OL)

9

Skills group meetings

Always prepare the skills group meetings!

- Do the exercise at home before the meeting
- Bring your own laptop to the meeting
- Discuss the exercise with your group members and the teaching assistant

10

Online Learning Material

1. [Canvas](#)
2. [Course manual](#) (with simple exercises)

Examination

1. **Final exam**
 - **last week of the block**
 - graded exam, 2.5 hours times
 - theoretical questions
 - few practical exercises/questions
2. **Assignments**
 - not graded, only pass/fail

Questions?